Characteristics of adsorption

1. Adsorption can occur in all interfacial surfaces i.e. the adsorption can occur in between gas-solid, liquid-solid, liquid-liquid, solid- solid and gas-liquid.
2. Adsorption is a spontaneous process and it is always accompanied by decrease in free energy. When ΔG reaches zero, the equilibrium is attained.

   We know, $\Delta G = \Delta H - T\,\Delta S$ where ΔG is Change in Free energy.

   ΔH is Change in enthalpy and ΔS = Change in entropy.
3. When molecules are adsorbed, there is always a decrease in randomness of the molecules.

   ie., $\Delta S<0$, and $T\Delta S$ is negative. Hence, adsorption is exothermic.

   Adsorption is a quick process whereas absorption is a slow process.

M.C. Bain introduced a term 'sorption' to represent the simultaneous adsorption and absorption. T. Graham used a term occlusion for sorption of gases on metal surfaces.

10.1.1 Types of adsorption

Adsorption is classified as physical adsorption and chemical adsorption, depending on the nature of forces acting between adsorbent and adsorbate. In chemical adsorption, gas molecules are held to the surface by formation of chemical bonds. Since strong bond is formed, nearly 400 KJ / mole is given out as heat of adsorption.

Examples

- Adsorption of O_2 on tungsten, Adsorption of H_2 on nickel, Adsorption of ethyl alcohol vapours on nickel.

In physical adsorption, physical forces such as van der waals force of attraction, dipole - dipole interaction, dispersion forces etc., exist between adsorbent and adsorbate. As these forces are weak, heat of adsorption is low, hence physical adsorption occurs at low temperatures.

Examples

(a) Adsorption of N_2 on mica.

(b) Adsorption of gases on charcoal.

The following table 10.1 illustrates the distinction between chemical and physical adsorption.

Table 10.1 Distinction between chemical and physical adsorption

Chemical adsorption or Chemisorption or Activated adsorption	Physical adsorption or van der waals adsorption or Physisorption
1. It is very slow	1. It is instantaneous
2. It is very specific depends on nature of adsorbent and adsorbate.	2. It is non-specific

XII U10-Surface Chemistry.indd 71 16-12-2022 19:40:12